Lab 3 - Routing Protocol

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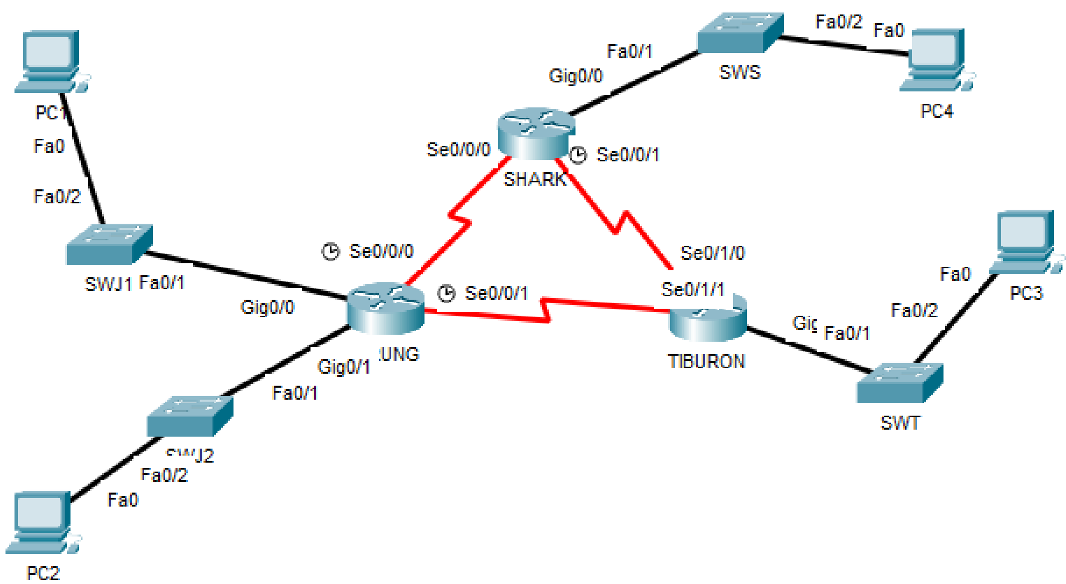
SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

Device	Subnetwork	Usable Hosts	Total IPs	Usable IPs	Prefix
JERUNG	LAN1	20	2^5 = 32	30	32 – 5 = /27
	LAN2	30	2^5 = 32	30	32 – 5 = /27
SHARK	LAN	60	2^6 = 64	62	32 – 6 = /26
TIBURON	LAN	50	2^6 = 64	62	32 – 6 = /26
Connections	JERUNG-SHARK	2	2^5 = 32	30	32 – 2 = /30
	JERUNG-TIBURON	2	2^2 = 4	2	32 – 2 = /30
	SHARK-TIBURON	2	2^2 = 4	2	32 – 2 = /30

Device	Subnetwork	Total IPs	Usable IPs
JERUNG	LAN1	172.18.110.0 – 172.18.110.31	172.18.110.1 – 172.18.0.30
	LAN2	172.18.110.32 – 172.18.110.63	172.18.110.33 – 172.18.0.62
SHARK	LAN	172.18.110.64 – 172.18.110.127	172.18.110.65 – 172.18.110.126
TIBURON	LAN	172.18.110.128 – 172.18.110.191	172.168.110.129 – 172.18.110.190
Connections	JERUNG-SHARK	172.18.110.192 – 172.18.110.195	172.18.110.193 – 172.18.110.194
	JERUNG-TIBURON	172.18.110.196 – 172.18.110.199	172.18.110.197 – 172.18.110.198
	SHARK-TIBURON	172.18.110.200 – 172.18.110.203	172.18.110.201 – 172.18.110.202

Device	Subnetwork	Prefix	Subnet Mask
JERUNG	LAN1	/27	11111111.11111111.11111111.11100000 = 255.255.255.224
	LAN2	/27	11111111.11111111.11111111.11111100 = 255.255.255.252
SHARK	LAN	/26	11111111.11111111.11111111.11000000 = 255.255.255.192
TIBURON	LAN	/26	11111111.11111111.11111111.11000000 = 255.255.255.192
Connections	JERUNG-SHARK	/30	11111111.11111111.11111111.11111100 = 255.255.255.252
	JERUNG-TIBURON	/30	11111111.11111111.11111111.11111100 = 255.255.255.252
	SHARK-TIBURON	/30	11111111.11111111.11111111.11111100 = 255.255.255.252

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
   a. CLI
      i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

   b. PT tool
      i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

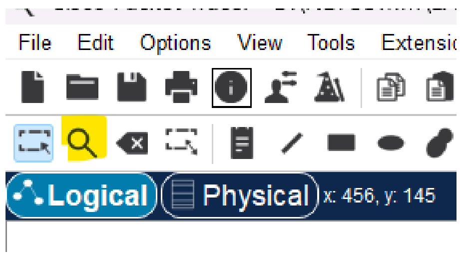


Figure B

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG

```
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1
```

SHARK

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
```

TIBURON

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
     172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.168.110.128/26 is directly connected, GigabitEthernet0/0
L       172.168.110.129/32 is directly connected, GigabitEthernet0/0
```

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time=1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.
   - PC1 and PC2 ping successfully because they are on the same subnet.
   - PC1 and PC4 are on different subnet. They cannot communicate directly because they are in different network, and connected to different routers. The router needs to forward the packets between the network.
   - PC1 does not know the path for PC4, and it shows Destination host unreachable.

5. What needs to be done to ensure all PCs can ping each other successfully?
   - Check the physical connectivity of the network is correct and proper
   - Verify the Serial Interface Configuration is correct
   - Check the neighbouring device configurations are correctly configurated.

Task 3 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

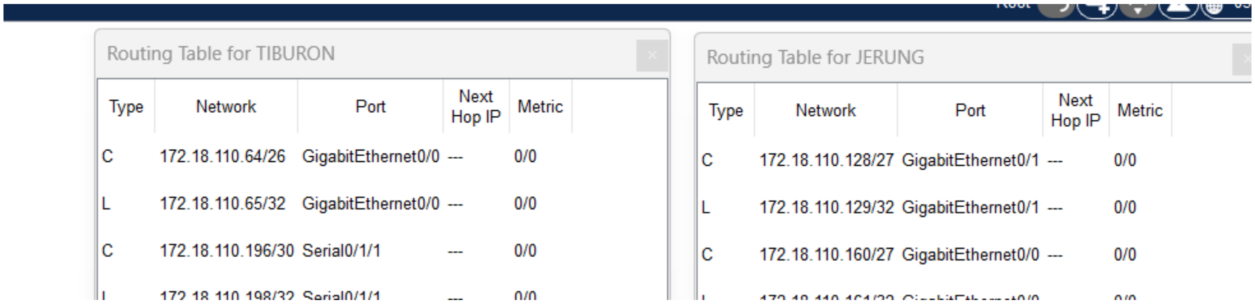


Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
```

Figure E

   a. What can you say about the addresses used in the 'network' instructions in Figure E?
   The 'network' instructions are used to include the 4 subnets to the router.
   RIP will advertise the subnets. It means that the router will exchange the routing information with other routers in the network.
   network 172.18.110.128 – include all interface within 172.18.110.128 subnet
   network 172.18.110.160 – include all interface within 172.18.110.160 subnet
   network 172.18.110.192 – include all interface within 172.18.110.192 subnet
   network 172.18.110.196 – include all interface within 172.18.110.196 subnet

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

a. What are the changes seen in TIBURON?

Addition of RIP route for subnet 172.18.110.0/27, 172.18.110.32/27, and 172.18.119.192/30. These routes are learned from another router via RIP and now is part of the TIBURON routing table. It will allow to forward traffic to these subnets.

b. What are the Networks with type R in TIBURON and JERUNG?

TIBURON – 172.18.110.0/27, 172.18.110.32/27, 172.18.119.192/30
JERUNG – 172.18.110.128/26, 172.18.110.200/30

c. Ping PC3 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=13ms TTL=126
Reply from 172.18.110.190: bytes=32 time=14ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=14ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 14ms, Average = 10ms
```

d. Ping PC4 from PC1. Was it successful?

No

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

e. Explain the reasons for your answer.

- PC1 and PC3 ping successfully because they are on the same subnet.
- PC1 and PC4 are on different subnet. They cannot communicate directly because they are in different network, and connected to different routers. The router needs to forward the packets between the network.
- PC1 does not know the path for PC4, and it shows Destination host unreachable.

f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.128
SHARK(config-router)#network 172.18.110.160
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.196
SHARK(config-router)#no auto-summary
SHARK(config-router)#exit
SHARK(config)#exit
SHARK#
%SYS-5-CONFIG_I: Configured from console by console
```

g. Open router SHARK's routing table and paste here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=12ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=11ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 12ms, Average = 6ms C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=16ms TTL=126 Reply from 172.18.110.30: bytes=32 time=18ms TTL=126 Reply from 172.18.110.30: bytes=32 time=14ms TTL=126 Reply from 172.18.110.30: bytes=32 time=14ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 14ms, Maximum = 18ms, Average = 15ms
2	C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=49ms TTL=126 Reply from 172.18.110.62: bytes=32 time=9ms TTL=126 Reply from 172.18.110.62: bytes=32 time=16ms TTL=126 Reply from 172.18.110.62: bytes=32 time=14ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 9ms, Maximum = 49ms, Average = 22ms C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=19ms TTL=126 Reply from 172.18.110.62: bytes=32 time=13ms TTL=126 Reply from 172.18.110.62: bytes=32 time=16ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 19ms, Average = 12ms
3	C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=19ms TTL=126 Reply from 172.18.110.190: bytes=32 time=14ms TTL=126 Reply from 172.18.110.190: bytes=32 time=18ms TTL=126 Reply from 172.18.110.190: bytes=32 time=8ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 8ms, Maximum = 19ms, Average = 14ms C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=13ms TTL=126 Reply from 172.18.110.190: bytes=32 time=16ms TTL=126 Reply from 172.18.110.190: bytes=32 time=14ms TTL=126 Reply from 172.18.110.190: bytes=32 time=16ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 13ms, Maximum = 16ms, Average = 14ms

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
    a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/24

192.168.1.1/24

	11000000 . 10101000 . 00000001 . 00000001	[IP address]
and	11111111 . 11111111 . 11111111 . 00000000	[Subnet mask]
	11000000 . 10101000 . 00000001 . 00000000	[Network Address]
	(192) . (168) . (1) . (0)	

    b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

    c. After this change, can PC4 and PC1 ping PC3?

No, because they are on different gateway.

```
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

e. Referring to the routing table, explain your findings.

TIBURON is directly connected to 192.168.1.0/24 subnet and route traffic for other subnets, including those in the 172.18.110.0/24.
SHARK is directly connected to 172.18.110.64/26, and has routing information for 172.18.110.0/27 and 172.18.110.32/27 subnets.
Both routers rely on RIP, which shows R (RIP-learn) in the routing table.

f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

Make sure the TIBURON have the RIP routes for other subnets.
Check the PCs have valid IP address, subnet masks and gateways.

g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.0
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#exit
TIBURON(config)#exit
TIBURON#
%SYS-5-CONFIG_I: Configured from console by console
```

h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

<table>
<tr><th>PC</th><th>Ping result</th></tr>
<tr><td>1</td><td>
<pre>
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=17ms TTL=126
Reply from 172.18.110.30: bytes=32 time=7ms TTL=126

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 17ms, Average = 6ms
</pre>
<pre>
C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=14ms TTL=126
Reply from 172.18.110.30: bytes=32 time=7ms TTL=126
Reply from 172.18.110.30: bytes=32 time=15ms TTL=126
Reply from 172.18.110.30: bytes=32 time=16ms TTL=126

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 16ms, Average = 13ms
</pre>
</td></tr>
<tr><td>2</td><td>
<pre>
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=17ms TTL=126
Reply from 172.18.110.62: bytes=32 time=7ms TTL=126
Reply from 172.18.110.62: bytes=32 time=6ms TTL=126
Reply from 172.18.110.62: bytes=32 time=15ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 17ms, Average = 11ms
</pre>
<pre>
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=14ms TTL=126
Reply from 172.18.110.62: bytes=32 time=10ms TTL=126
Reply from 172.18.110.62: bytes=32 time=33ms TTL=126
Reply from 172.18.110.62: bytes=32 time=11ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 33ms, Average = 17ms
</pre>
</td></tr>
<tr><td>4</td><td>
<pre>
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=13ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=16ms TTL=126
Reply from 172.18.110.126: bytes=32 time=10ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 10ms
</pre>
<pre>
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=17ms TTL=126
Reply from 172.18.110.126: bytes=32 time=12ms TTL=126
Reply from 172.18.110.126: bytes=32 time=7ms TTL=126
Reply from 172.18.110.126: bytes=32 time=16ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 17ms, Average = 13ms
</pre>
</td></tr>
</table>

REFLECTION

What have you learned in this task?

I learnt how to set up and configurate IP addresses, subnet masks, and routing on the routers. I also learnt how to use RIP (Routing Information Protocol) to help router communicates with each other and share router so that device in different subnet can communicate with each other. I also saw the importance of setting correct gateway on PC so that they can reach other device, and how change in router IP addresses can affect the whole network. Finally, I learnt how to read routing tables and use the tools like CLI, or to ping to check if everything is working fine.